

Unit 01 :- Fundamentals of Computer Network

* Computer network:-

• Definition:-

A computer network is the interconnection of two or more computers to exchange data & to share resources with each other.

ex- it may connect computers, printers, scanners etc.

• Need :-

i) To share computer files -

Users can share files with each other. For ex- in a company, one file can be shared & used by its all branches.

ii) To share computer equipment -

Network enables users to share equipments such as laser printers & large hard-disk drives by networking microcomputers or workstations together.

iii) To enable unlike computer equipment to communicate -

A company with computers of multiple uses using several OS, including MS-DOS, windows, UNIX, can't share files from one computer to another unless arranged using ~~net~~ network OS.

iv) To improve communication speed & accuracy -
messages can be sent fastly & accurately.

v) To reduce the cost of data transfer -
The cost of transferes of files using computers in network is less expensive.

vi) High Reliability -
Different copies of same files are available.

• Applications :-

- i) sharing of resources such as printers
- ii) Sharing of expensive software's & database
- iii) communication from one computer to another
- iv) Exchange of data & information among users via network.
- v) Sharing of information over geographically wide areas.

• Component of computer network:-

- i) Two or more computers
- ii) cables as links between the computers
- iii) A network interfacing card (NIC) on each computer
- iv) switches
- v) Software called operating system (OS)

1.2] Network Benefits :-

• File Sharing -

You can easily share data between different users, or access it remotely if you keep it on other connected devices.

• Resource sharing -

Using network connected peripheral devices like printers, scanners and copiers or sharing software between multiple users, saves money.

• Maintaining the network -

A well-maintained network ensures that your connectivity is optimized thus it is easy for users to access data & other resources without too much waiting.

• Backing up data -

Network backup is essential to ensure the security of your business & all of its assets.

• Sharing a single internet connection -

It is cost efficient & can help protect your systems if you properly secure the network.

- Increasing storage capacity-

You can access files and multimedia, such as images & music, which you store remotely on other machines or network attached storage devices.

- Improved communication-

Networking computers can also help you improve communication so that users can get & share information easily.

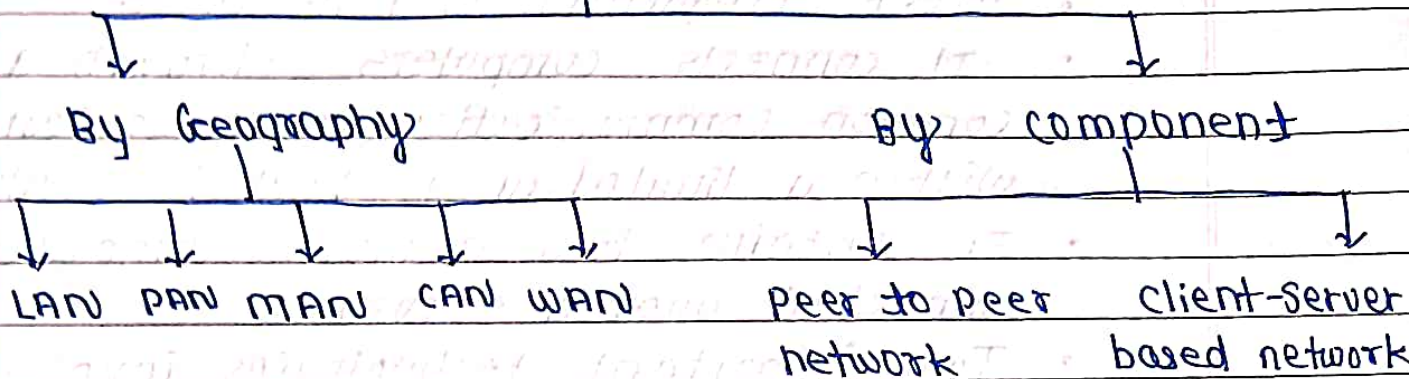
- Cost benefits :-

storing information in one centralized database can also help to reduce costs & drive efficiency.

- * Disadvantages of networks :-

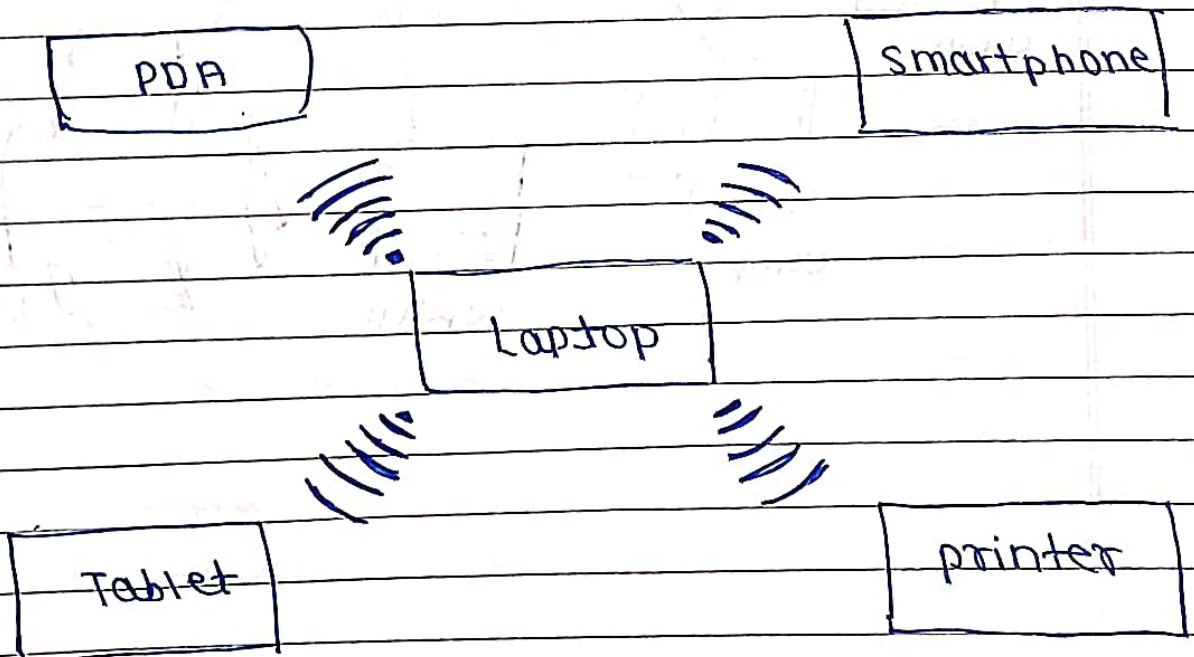
- i) High cost of installation
- ii) Require time for administration
- iii) Failure of server
- iv) cable faults

* Computer network classification



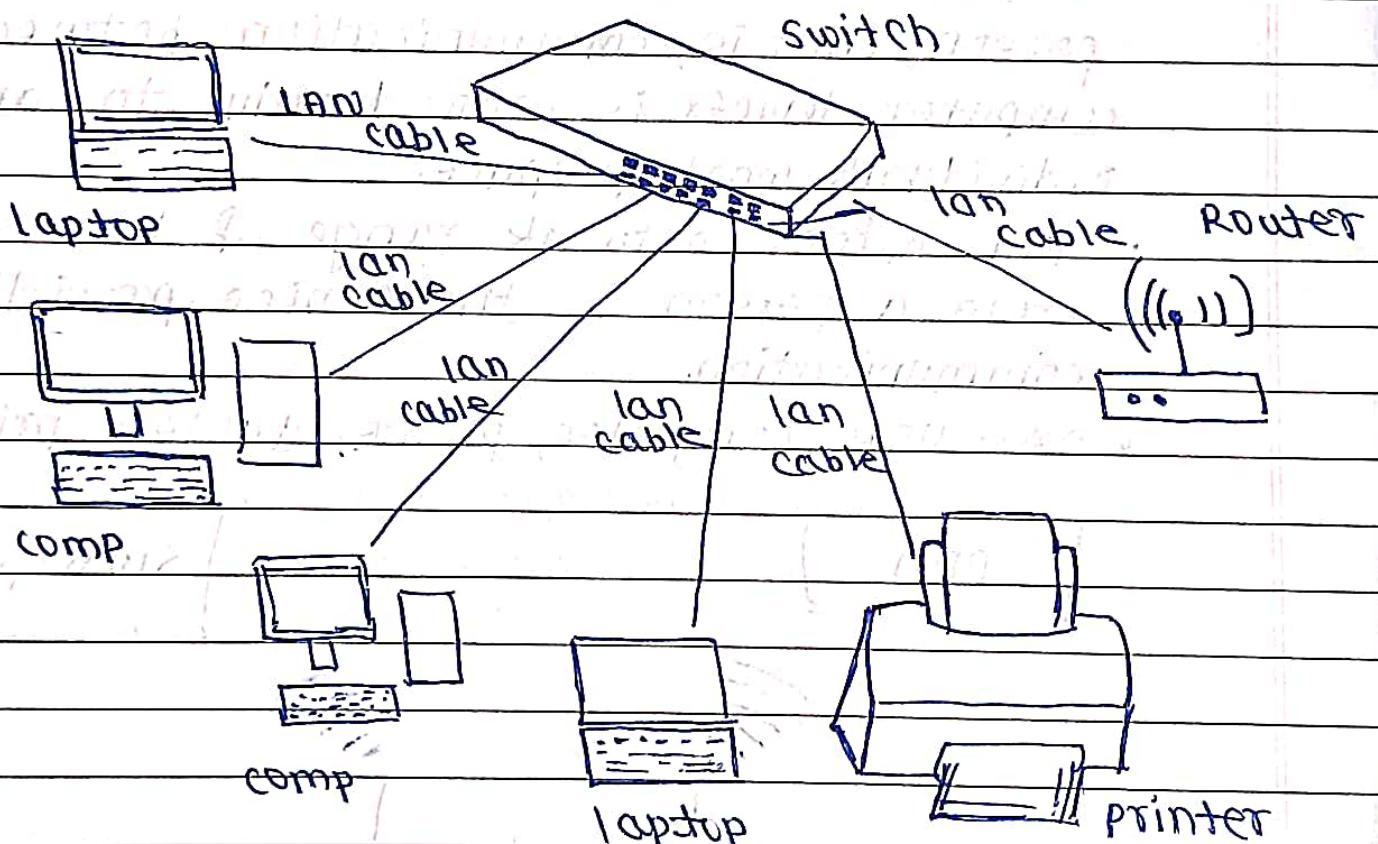
1] Personal Area Network (PAN) :-

- most basic type of computer network
- This network is restrained to a single person, that is, communication between the computer devices is centred only to an individual's work space.
- PAN offers network range of 10 meters from a person to the device providing communication.
- ex - USB, computer, phone, tablet, printer etc.



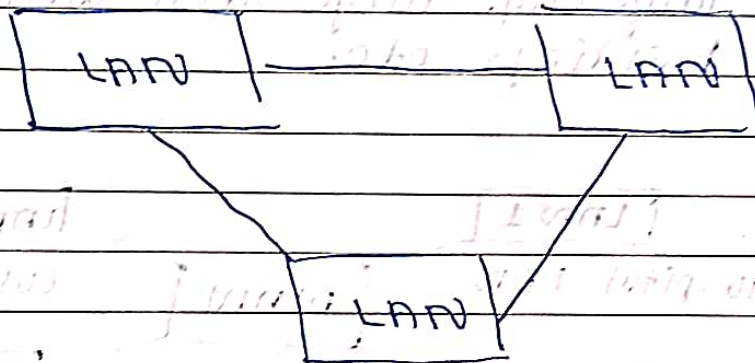
2] Local Area Network (LAN):

- most frequently used network.
- It connects computers through a common communication path, contained within a limited area, that is locally.
- It contains two or more computers connected over a server.
- Two important technologies involved in this network are Ethernet & wi-fi.
- Ex - networking in home, school, library, college, office etc.



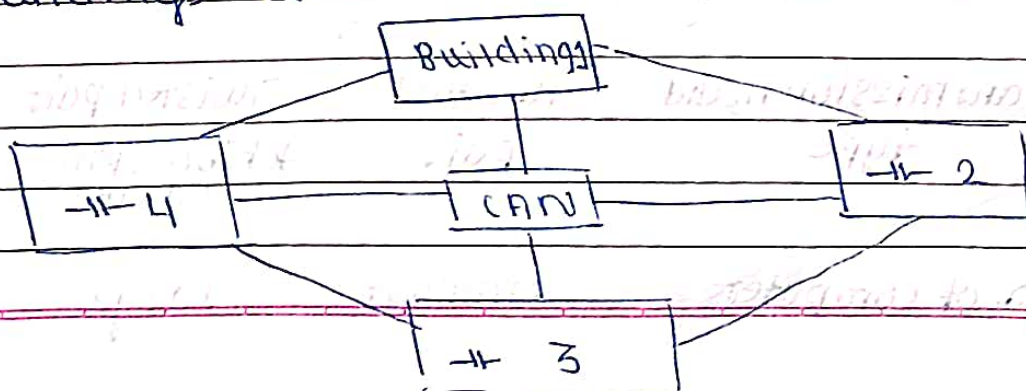
3] Wide Area Network (WAN) :-

- It connects computers over a large geographical distance through a shared communication path.
- It extends over many locations.
- WAN is group of local area networks that communicate with each other.
- ex- Internet.



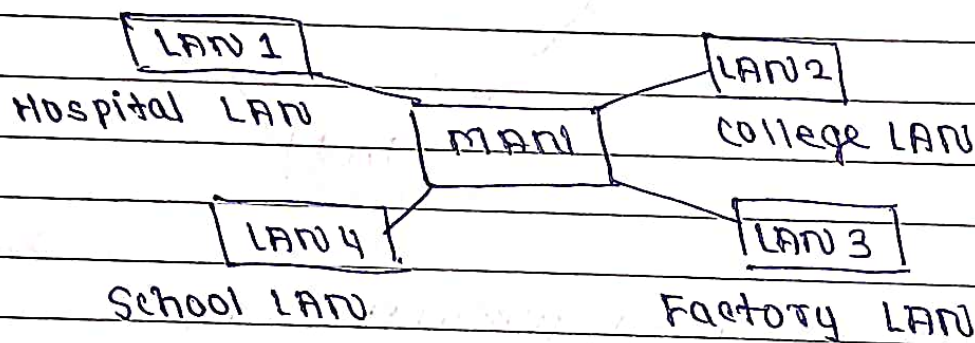
4] Campus Area Network (CAN) :-

- CAN is bigger than LAN but smaller than MAN.
- usually used in places like a school or college.
- It covers limited geographical area that is, it spreads across several buildings within the campus.
- ex- networks that covers school, colleges, buildings etc.



6] Metropolitan Area network (MAN) :-

- MAN is larger than LAN but smaller than WAN.
- It connects computers over a geographical distance through a shared communication path over a city, town or metropolitan area.
- Ex - networking in towns, cities, a single large city, large area within multiple buildings etc.



• Comparison :-

Criteria	LAN	MAN	WAN
Cost	Low	High	Higher
Network size	Small	larger	largest
Speed	Fastest	slower	slowest
Transmission media type	Twisted pair	Twisted pair & Fibre optic cables	Fibre optic, radio wave & satellite
no. of computers	smallest	large	largest

1.4] Classification of network by their component role :-

- i) Peer to Peer network
- ii) Client server based network

* Peer to Peer network :-

• A peer to peer network is analogous to a company that uses decentralized management where decisions are made locally.

• It is a network with all nodes acting as both servers and clients.

• A PC can access files located on another PC & can also provide files to other PC.

• All computers in peer to peer network has equal responsibilities & capabilities to use the resources available on network.

• It is useful for small network, when each computer maintains its own account & their security settings.

• Windows 2000 is P to P OS.

• P to P network don't have central control system thus it doesn't have central login process so it has very less security.

- we can use p to p network in following working condition:-

- i) IF network security is not an important issue.
- ii) no. of users is less than 10.
- iii) All users, are situated in small area.
- iv) IF possibility of future expansion is less.

- Advantages :-

- i) no extra investment in server hardware or software.
- ii) no need of server computer.
- iii) Easy to administer.
- iv) no n/os (network operating system) is required.
- v) Easy setup & lower cost.
- vi) User can control resource sharing.
- vii) User is not dependent on other computers for its operations.